

Bowel Habit Disorders

Presented by

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Agenda

- Diarrhea
- Constipation
- Malabsorption Syndrome
- Celiac Disease



Diarrhea



Definition :

frequent loose stool $> 300\text{gm} / \text{day}$.

[false diarrhea] frequent stool $< 300\text{gm/day}$
usually due to $\uparrow$ intestinal motility.



Mechanisms:

- Osmotic diarrhea: large non absorbed hypertonic substances in GIT lumen => ↑ fluid accumulation.
- Secretory diarrhea: results from active secretion of intestinal fluid and electrolytes and decrease absorption.
 - 1- Enterotoxins e.g. cholera and E-coli .
 - 2- Hormones as vasoactive intestinal polypeptide secreting tumours(VIPomas), gastrin and glucagon.



- Inflammatory diarrhea: Due to damage of int.mucosa => loss of fluid and blood and ↓ absorption of fluid and electrolytes as in :
 - IBD as crohn's and ulcerative colitis
 - Infection as shigellosis
- Abnormal intestinal motility: false diarrhea [↑ frequency but no ↑ in volume] as in
 - 1.DM.
 - 2.Hyperthyroidism.
 - 3.Post vagotomy.



Aetiology



Acute diarrhea:

- Dietary indiscretions (حماقة) and unripe fruits.
- Infections as food poisoning, viral GE, cholera.
- Traveller,s diarrhea: E. coli, shigella or E. histolitica.



Chronic diarrhea:

- Parasitic, viral or Fungal infection
- IBD and IBS
- Malabsorption Gut resection
- Drugs
- Colonic tumours
- Endocrinal diarrhea:
 - VIPomas => more than 3 L/day and ↑ level of VIP. (pancreatic cholera)
 - Gastrinoma => ↑ gastrin level in ZE syndrome
 - Glucagonoma => ↑ glucagon level and ↑ blood glucose level.
 - Carcinoid syndrome => ↑ urinary [5-HIAA].
 - Medullary carcinoma of thyroid => ↑ level of calcitonin.
- Purgative abuse.
- Chronic idiopathic (pseudo-pancreatic cholera)



Clinical picture



- Acute diarrhea:
 - Fever, abd. Pain and vomiting usually present.
 - Dehydration may occur in sever cases.
- Chronic diarrhea:
 - CP of cause.



Investigations



Acute diarrhea:

Necessary if diarrhea prolonged > one week

- Stool examination for ova, cysts, pus and parasites.
- Sigmoidoscopy, rectal biopsy may (rarely needed).

Chronic diarrhea:

- Investigations of the cause.
- Sigmoidoscopy, rectal biopsy.



Treatment



Acute diarrhea:

- Fluid replacement enteral or parenteral according to the condition.
- Antibiotics for bacterial diarrhea.
- Antidiarrheal drugs may impair clearance of responsible pathogens as loperamide or codeine phosphate.
- Chronic diarrhea: treatment of the cause.



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Definition

- Defecation twice a week or less.

Normal bowel habit vary from:

2-3 times / day to once every 2-3 days.



Aetiology:

- Physiological: Immobility, elderly, pregnancy or ↓ fiber intake
- Intestinal obstruction as in tumours, obstructive hernias or volvulus.
- Colonic disorders as IBS or megacolon.
- Pain as in anal fissure or thrombosed haemorrhoids.



- Drugs as anticholinergics, iron, opiates calcium channel blockers or diuretics.
- Immobility.
- Endocrinal and metabolic : diabetes mellitus, hypothyroidism, porphyria and hypercalcemia



Investigations:

- investigations for the cause.
- Sigmoidoscopy or colonoscopy and biopsy.
- Ano-rectal manometry may be done.



Treatment

- Treatment of cause.
- Bulk forming laxatives [high fibre diet as fruits and vegetables contain non absorbable polysaccharides or methylcellulose which form a bulk and st. int. motility]
- Stimulant laxatives as bisacodyl or docusate sodium
- Osmotic laxatives as lactulose or magnesium sulphate



- Enema [docusate sodium or sodium citrate] may be needed in some cases.
- Suppositories as glycerol or bisacodyl
- Prokinetic agents: as mosapride, itopride, domperidone or metoclopramide.



Recent ttt



- Prucalopride:

Stimulate serotonin 5-HT₄ receptor ==> stimulates colonic mass movements, which provide the main propulsive force for defecation.



- Lubiprostone:
a bicyclic fatty acid that acts by specifically activating **chloride channels** on the apical aspect of GIT epithelial cells ==> producing a **chloride-rich fluid secretion** ==> softening of stool, increase motility, and promote spontaneous bowel movements.

